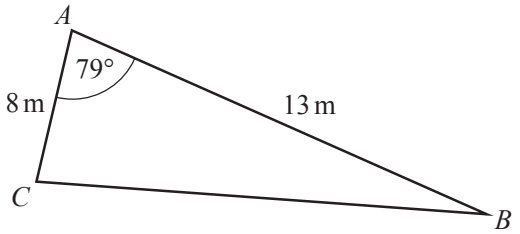
6 (a)



NOT TO SCALE

The diagram shows triangle *ABC*.

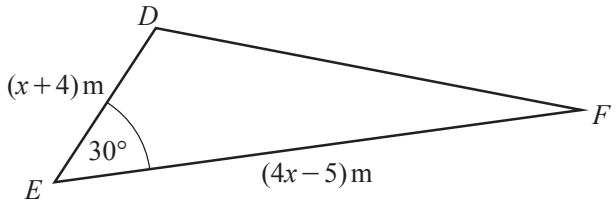
(i) Use the cosine rule to calculate *BC*.

BC = ... m [4]

(ii) Use the sine rule to calculate angle *ACB*.

Angle *ACB* = ... [3]

(b)



NOT TO SCALE

The area of triangle *DEF* is $70\,\text{m}^2$.

(i) Show that $4x^2 + 11x - 300 = 0$.

[4]

(ii) Use the quadratic formula to solve $4x^2 + 11x - 300 = 0$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... or $x =$ ..... [4]

(iii) Find the length of *DE*.

DE = ... m [1]